

AgriVision BD: A Multi-Crop Leaf Disease Detection System for Edge-Device Deployment in Bangladesh

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Abstract

Agricultural productivity in Bangladesh is severely impacted by crop diseases that go undetected due to limited access to expert agronomists, especially in rural areas. This report presents **AgriVision BD**, a lightweight multi-crop leaf-disease detection system designed for on-device inference on Android smartphones and edge hardware. We curate and merge disease image datasets across five staple crops grown in Bangladesh — **Brassica** (cabbage/cauliflower), **Rice**, **Corn**, **Potato**, and **Wheat** — retaining only disease classes relevant to the Bangladeshi agricultural context. We train and evaluate both lightweight convolutional classifiers and the modern **YOLO26** family of models. The best-performing models achieve test-set top-1 accuracies of up to **97.29%** (Brassica), **96.49%** (Rice), **97.23%** (Corn), and **95.04%** (Potato), with all weights exported in TFLite/ONNX format for edge deployment. This report documents dataset provenance, training configurations, per-class evaluation metrics, and comparative analysis across all trained models.

Introduction

Bangladesh is an agrarian economy where rice, wheat, potato, corn, and brassica vegetables constitute the primary food crops. Disease outbreaks — caused by fungal, bacterial, and viral pathogens — can destroy 20–40% of yields if left undetected [1]. Traditional pest-management relies on visual inspection by trained agronomists, a resource not uniformly available in rural Bangladesh.

Recent advances in deep learning, particularly compact convolutional neural networks (CNNs) and transformer-hybrid architectures, have enabled high-accuracy disease recognition from leaf photographs. However, most existing benchmark datasets are curated for temperate climates (North America, Europe) and include disease classes that do not occur in Bangladesh.

Our contributions:

- We aggregate multiple public datasets per crop, filter to retain only classes confirmed to affect Bangladesh, and publish the merged splits on Kaggle.
- We fine-tune and rigorously evaluate five classifiers covering five crops (26 total disease + healthy classes).
- We export every model in ONNX (FP16) and TFLite (INT8/FP16) formats, enabling real-time inference (<30 ms/image) on off-the-shelf Android devices.

Dataset Construction & Methodology

Source Datasets and Filtering

Each crop dataset was assembled from multiple publicly available repositories (Mendeley Data and Kaggle), then filtered to retain only disease classes clinically relevant to Bangladesh. Table 1 summarises the original sources.

Table 1: Dataset sources per crop (URLs in § 6).

Crop	Primary Sources	Platform
Brassica	AgriVision-Brassica (custom)	Kaggle / Mendeley
Rice	IRRI Rice Leaf Disease + others	Mendeley + Kaggle
Corn	Corn/Maize Leaf Disease	Kaggle + Mendeley
Potato	PlantVillage + BD-specific	Mendeley + Kaggle
Wheat	Wheat Plant Diseases (6-cls)	Mendeley + Kaggle

Class Selection for Bangladeshi Context

A literature review of crop disease surveys in Bangladesh [2] guided the inclusion/exclusion of disease classes. We excluded diseases with no confirmed occurrence in Bangladesh (e.g., Stewart’s Wilt for corn in temperate zones) and retained the following classes:

- **Brassica (11 classes):** Cabbage Alternaria Spot, Cabbage Black Rot, Cabbage Downy Mildew, Cabbage Healthy, Cauliflower Alternaria Disease, Cauliflower Bacterial Soft Rot, Cauliflower Bacterial Spot, Cauliflower Black Spot, Cauliflower Downy Mildew, Cauliflower Healthy, Cauliflower Nutrient Deficiency.
- **Rice (8 classes):** Bacterial Leaf Blight, Brown Spot, Healthy, Leaf Blast, Leaf Scald, Narrow Brown Leaf Spot, Rice Hispa, Sheath Blight.
- **Corn (4 classes):** Common Rust, Northern Leaf Blight, Gray Leaf Spot, Healthy.
- **Potato (3 classes):** Early Blight, Healthy Leaf, Late Blight.
- **Wheat (6 classes):** Leaf Rust, Stem Rust, Stripe Rust, Blast, Powdery Mildew, Healthy.

Dataset Statistics

Table 2 provides the final dataset split statistics after merging and filtering.

Table 2: Dataset split sizes after filtering.

Crop	Classes	Train	Val	Test	Total
Brassica	11	2,732	426	443	~3,600
Rice	8	~3,000	—	~342	~3,400
Corn	4	4,373	936	940	6,250
Potato	3	7,600	855	1,028	9,308
Wheat	6	4,839	120	300	5,259
Total	32	~22,544			~27,817

Training Setup

All models were trained on Kaggle’s free GPU tier (Tesla T4 / P100). Two model families were explored:

(a) **YOLO26-cls**: Ultralytics’ 2026 release of the YOLO classification variant. We used `yolo26s-cls` for Brassica, Corn, and Potato and `yolo26n-cls` (nano) for Rice. Key hyperparameters: 100 training epochs, cosine LR scheduler, early stopping (patience 15–20), data augmentation (HSV jitter, random flip, mosaic, mixup), AMP (automatic mixed precision).

(b) **BrassicaHybridNet (EfficientNet-B1 + ViT SE)**: A custom hybrid architecture combining a MobileNetV3-based backbone with Squeeze-and-Excitation attention blocks and a lightweight transformer module, developed for the Brassica crop. This is compared against the YOLO26s baseline on the same dataset.

Model Details and Evaluation

Brassica (Cabbage & Cauliflower) — 11 Classes

Model A — YOLO26s-cls: Trained on the AgriVision-Brassica dataset (240×240 resolution, 11 classes). Best checkpoint selected by validation Top-1. Test Top-1: **97.29%** (PyTorch); ONNX evaluation on 410 images: **96.59%**. Model size: 11.05 MB (.pt), 21.84 MB ONNX.

Model B — BrassicaHybridNet (EfficientNet-B1 + ViT SE): A custom hybrid architecture (7.3M parameters). Trained for 80 epochs with phase-based LR scheduling. Best validation accuracy: **97.42%** (epoch 61). Test accuracy: **93.45%**. Exported as TFLite Float16 (13.98 MB). Original README reported best val accuracy of 87.32% from an earlier training run; the full extended run reached 97.42% validation accuracy.

Table 3 shows YOLO26s per-class performance on the test set.

Table 3: Brassica YOLO26s per-class metrics (test set, 410 images).

Class	Prec.	Rec.	F1	N
Cabbage Alternaria Spot	0.818	0.750	0.783	12
Cabbage Black Rot	0.857	0.857	0.857	7
Cabbage Downy Mildew	0.400	0.667	0.500	3
Cabbage Healthy	0.727	0.800	0.762	10
Cauliflower Alt. Disease	0.981	1.000	0.991	53
Cauliflower Bact. Soft Rot	1.000	0.969	0.984	32
Cauliflower Bact. Spot	0.984	1.000	0.992	61
Cauliflower Black Spot	1.000	1.000	1.000	39
Cauliflower Downy Mildew	0.984	0.968	0.976	63
Cauliflower Healthy	1.000	0.986	0.993	139
Cauliflower Nutrient Def.	1.000	1.000	1.000	24
Weighted Avg	0.966	0.966	0.965	443

Note: Cabbage classes (especially Downy Mildew, only 3 test samples) show lower performance. Cauliflower classes achieve near-perfect metrics across the board. Imbalance between cabbage (small sample) and cauliflower (large sample) is a key limitation.

Rice — 8 Classes

Model: YOLO26n-cls (nano, 3.2 MB .pt, 6.19 MB ONNX). Input: 320×320. Trained for the 8-class rice disease dataset. Test Top-1 accuracy: **96.49%** (standard inference; TTA confirms same result). Inference speed: 1.1 ms/image on T4 GPU.

Table 4: Rice YOLO26n per-class metrics (test set).

Class	Prec.	Rec.	F1	N
Bacterial Leaf Blight	0.957	0.957	0.957	47
Brown Spot	0.953	1.000	0.976	41
Healthy Leaf	1.000	1.000	1.000	42
Leaf Blast	0.952	0.851	0.899	47
Leaf Scald	0.952	0.976	0.964	41
Narrow Brown Leaf Spot	0.905	1.000	0.950	38
Rice Hispa	1.000	1.000	1.000	42
Sheath Blight	1.000	0.955	0.977	44
Weighted Avg	0.966	0.965	0.965	342

Note: Leaf Blast is the hardest class (F1 = 0.899), likely due to visual similarity with Narrow Brown Leaf Spot. The model is remarkably compact (3.2 MB) while maintaining high accuracy, making it ideal for edge deployment.

Corn (Maize) — 4 Classes

Model: YOLO26s-cls. Input: 240×240 (padded to 256 for stride alignment). Classes: Common Rust, Northern Leaf Blight, Gray Leaf Spot, Healthy. Training: 100 epochs, patience 15, batch 32. Test Top-1: **97.23%**. Model size: 11.0 MB (.pt), 21.8 MB ONNX.

Table 5: Corn YOLO26s per-class metrics (test set, 940 images).

Class	Prec.	Rec.	F1	N
Common Rust	0.977	0.991	0.984	213
Northern Leaf Blight	0.955	0.950	0.953	222
Gray Leaf Spot	0.997	0.988	0.993	333
Healthy	0.942	0.948	0.945	172
Overall	0.972	0.972	0.972	940

Note: Gray Leaf Spot achieves the highest F1 among all corn classes. Northern Leaf Blight and Healthy classes show slight confusion with each other, which is expected as both may appear as pale/yellowish leaf regions.

Potato — 3 Classes

Model: YOLO26s-cls. Input: 224×224, trained for 59 epochs (early stopping at patience 20). Dataset: 9,308 images. Test-set metrics are the most thoroughly documented across all models.

Table 6: Potato YOLO26s test-set summary.

Metric	Value
Top-1 Accuracy	95.04%
Balanced Accuracy	96.27%
Weighted Precision	95.17%
Weighted Recall	95.04%
Weighted F1-Score	95.05%
Macro F1-Score	94.71%
Matthews Corr. Coef.	0.9176
Cohen’s Kappa	0.9173
AUC — Early Blight	0.9943
AUC — Healthy Leaf	0.9990
AUC — Late Blight	0.9918
PyTorch .pt size	11.0 MB
TFLite INT8	5.6 MB
ONNX FP16	10.9 MB
Inference (PyTorch)	25.8 ms/img
Inference (TFLite INT8)	24.1 ms/img
Inference (ONNX FP16)	11.1 ms/img

The Potato model’s MCC of 0.9176 and Cohen’s Kappa of 0.9173 confirm excellent agreement beyond chance, making the results statistically robust for a 3-class problem.

Wheat — 6 Classes

Model: YOLO26n-cls. Input: 320×320. Classes: Leaf Rust, Stem Rust, Stripe Rust, Blast, Powdery Mildew, Healthy. Trained for 100 epochs. Test Top-1 accuracy: **85.67%**. Test Top-5: **100.00%**. Model size: 3.2 MB (.pt), 6.2 MB ONNX, 1.6 MB TFLite. Dataset statistics:

Table 7: Wheat dataset split (model pending).

Class	Train	Val	Test
Leaf Rust	1,301	20	50
Stem Rust	234	20	50
Stripe Rust	576	20	50
Blast	647	20	50
Powdery Mildew	1,081	20	50
Healthy	1,000	20	50
Total	4,839	120	300

Cross-Model Comparison

Table 8 summarises the top-1 test accuracy and model size across all trained models. Figure-level comparison is presented in terms of four key dimensions: accuracy, model compactness, number of classes, and dataset size.

Table 8: Cross-model performance comparison.

Crop	Model	Cls.	Test Top-1	Param.	.pt (MB)	TFLite
Brassica	YOLO26s-cls	11	97.29%	~5.4M	11.1	N/A*
Brassica	HybridNet (EffB1+ViT)	11	93.45%	7.3M	28.3	14.0 MB
Rice	YOLO26n-cls	8	96.49%	~1.9M	3.2	N/A**
Corn	YOLO26s-cls	4	97.23%	~5.4M	11.0	Exported
Potato	YOLO26s-cls	3	95.04%	~5.4M	11.0	5.6 MB (INT8)
Wheat	YOLO26n-cls	6	85.67%	~1.9M	3.2	1.6 MB

*TFLite export failed due to Protobuf version conflict (to be fixed). **ONNX exported (6.2 MB).

Key observations:

1. **YOLO26s-cls consistently outperforms** the custom HybridNet on Brassica (97.29% vs. 93.45%), while also being more compact (11 MB vs. 28 MB PyTorch weights).
2. **YOLO26n-cls (nano)** achieves 96.49% on Rice with only 3.2 MB — the smallest model in the suite, ideal for highly resource-constrained deployment.
3. **Potato** models are the most rigorously evaluated (MCC, Kappa, AUC, TFLite INT8 benchmarked at 24.1 ms/image) and exceed the project target of ≥95% accuracy.
4. **Corn** achieves 97.23% on a challenging 4-class problem with 940 test images; ONNX FP16 export is available.
5. **Cabbage classes** consistently underperform relative to Cauliflower classes in the Brassica dataset due to severe class imbalance (cabbage: 3–12 test samples vs. cauliflower: 24–139).

Edge Deployment Pipeline

All trained models target deployment on Android smartphones via the following export pipeline:

1. Train with Ultralytics YOLO26 (PyTorch, GPU) or custom PyTorch.
2. Export to ONNX (FP16, opset 12) for cross-platform server/Android inference via ONNX Runtime Mobile.
3. Export to TFLite (INT8 / FP16) for integration with **react-native-fast-tflite** or TensorFlow Lite Android API.

4. Validate exported model: confirm output shape matches ($1 \times C$ softmax logits), apply threshold ($p_{max} < 0.60 \Rightarrow$ “retake photo”).

Table 9 summarises available deployment artifacts.

Table 9: Deployment artifact availability per crop.

Crop	Model	ONNX	TFLite INT8	TFLite FP16
Brassica	YOLO26s	✓ (21.8 MB)	×	×
Brassica	HybridNet	✓ (27.8 MB)	×	✓ (14.0 MB)
Rice	YOLO26n	✓ (6.2 MB)	×	×
Corn	YOLO26s	✓ (21.8 MB)	✓	✓
Potato	YOLO26s	✓ (10.9 MB)	✓ (5.6 MB)	—
Wheat	YOLO26s	✓ (6.2 MB)	—	✓ (1.6 MB)

Dataset Links

Rice:

- <https://data.mendeley.com/datasets/vwv3nry3wr/1>
- <https://kaggle.com/datasets/dedeikhsandwisaputra/rice-leafs-disease-dataset>

Wheat:

- <https://data.mendeley.com/datasets/5gc7hwydwg/1>
- <https://kaggle.com/datasets/kushagra3204/wheat-plant-diseases>

Potato:

- <https://data.mendeley.com/datasets/ptz377bwb8/1>
- <https://data.mendeley.com/datasets/pmbc875pr7/1>
- <https://kaggle.com/datasets/aarishasifkhan/plantvillage-potato-disease-dataset>

Corn:

- <https://kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf-disease-dataset>
- <https://data.mendeley.com/datasets/vy629dngm8/1>

Brassica (custom AgriVision split — Kaggle):

- <https://kaggle.com/datasets/khanraiyan/agrivision-brassica>

Limitations and Future Work

- **Class imbalance:** Cabbage sub-classes are severely under-represented ($n \leq 12$ test images). We plan SMOTE/oversampling and weighted loss functions.
- **TFLite export failures:** Protobuf version conflicts on Kaggle prevented TFLite INT8 export for Brassica and Rice YOLO models. Will be resolved in a local environment.
- **Wheat model:** The baseline model achieved a modest Top-1 accuracy of 85.67%, suggesting that hyperparameter tuning, class-weight balancing, or a larger model architecture is required to meet the 95% target.
- **Real-world validation:** No field images from Bangladeshi farms have been tested. A field data collection campaign is planned with BADC (Bangladesh

Agricultural Development Corporation) collaboration.

- **Multi-crop one-model:** Future work will explore a unified multi-head model to reduce the app’s model size footprint.
- **Bengali language UI:** The mobile app will feature a fully Bangla-language interface for farmer accessibility.

Conclusion

This report presents AgriVision BD, a comprehensive crop disease detection system targeting edge-device deployment for Bangladeshi farmers. We demonstrate that the YOLO26s-cls architecture consistently achieves $\geq 95\%$ test accuracy across Brassica, Corn, and Potato crops, while the ultra-compact YOLO26n-cls delivers near-equivalent performance on Rice at only 3.2 MB. All models are exported in ONNX and/or TFLite format and validated for mobile inference. The primary challenge remains data scarcity for certain crop/disease combinations in the Bangladeshi context, which will be addressed through future field data collection. The combination of curated Bangladesh-specific datasets, high-accuracy lightweight models, and an open-source mobile deployment pipeline positions AgriVision BD as a practical, deployable solution for early crop disease detection in rural Bangladesh.

References

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